Introduction to Quantum Physics and Structure of Matter

Lecture 6

October 24, 2026

TU/e EINDHOVEN UNIVERSITY OF TECHNOLOGY

Course Contents

Lecture	Date	Topics	Material
1	03-09-2025	•Relativistic dynamics •Review electromagnetic waves •Photoelectric effect •Blackbody radiation	Book: Ch 2.1-2.5, 2.7, 3.1-3.3
2	10-09-2025	•Compton scattering •De Broglie's hypothesis •Experimental evidence for De Broglie waves •Uncertainty relationships for classical particles •Heisenberg uncertainty relationships	Book: Ch. 3.4, 4.1-4.4
3	17-09-2025	•Behaviour of a wave at a boundary •Confining a particle •The Schrödinger equation •Applications of the Schrödinger equation	Book: Ch. 5.1-5.4
4	24-09-2025	•Basic properties of atoms •Scattering experiments and the Thomson model •The Rutherford nuclear atom •Line spectra •The Bohr model •The Franck-Hertz experiment •Deficiencies of the Bohr model	Book: Ch. 6.1-6.6, 6.8

Course Contents

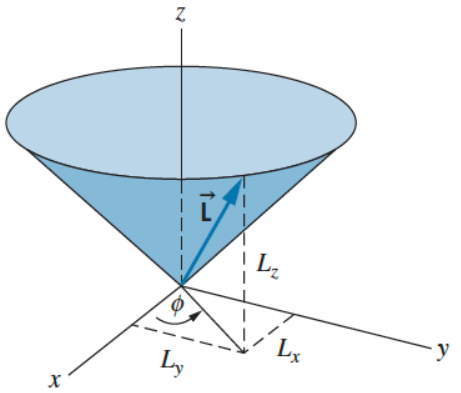
Lecture	Date	Topics	Material
5	01-10-2025	•A one-dimensional atom •Angular momentum in the hydrogen atom •The hydrogen atom wave functions •Radial probability densities •Angular probability densities	Book: Ch. 7.1-7.7
6	**08-10-2025**	**•The Pauli exclusion principle** **•Electron states in many-electron atoms** **•Outer electrons: screening and optical transitions** **•Inner electrons: absorption edges and X-rays**	**Book: Ch. 8.1-8.3, 8.5**
7	15-10-2025	•Nuclear constituents •Nuclear sizes and shapes •Alpha decay •Beta decay •Gamma decay and nuclear excited states	Book: Ch. 12.1-12.3 12.6-12.9
8	t.b.d.	Visit to the Catharina hospital, radiotherapy department	
Final Exam: Thursday October 30th, 13:30-16:30			

Recap Previous Lecture

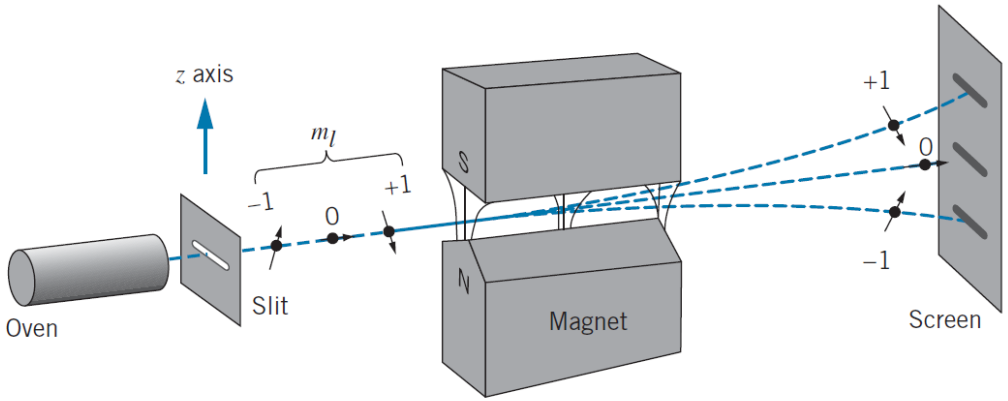
- Schrödinger equation for H-atom
- Quantum angular momentum

$$\left|\vec{L}\right| = \sqrt{l(l+1)}\hbar$$

$$L_z = m_l\hbar$$



- Stern-Gerlach experiment

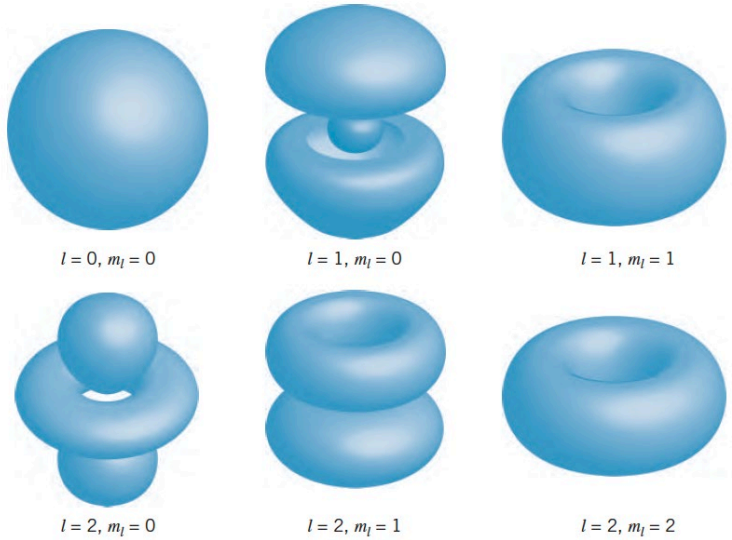


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Recap Previous Lecture

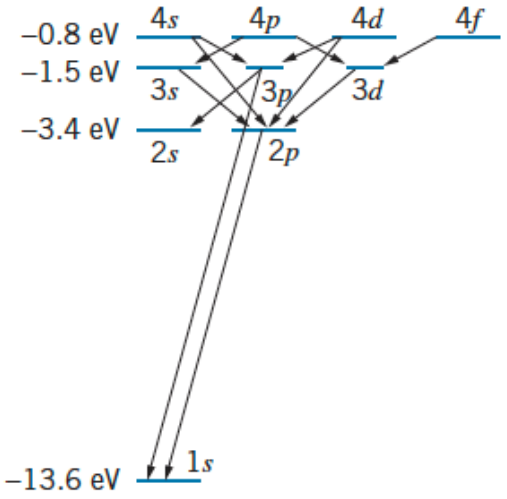
- Quantum numbers

n	principal quantum number,	1,2,3, …
l	angular momentum quantum number,	
	magnetic quantum number,	$0, \pm 1, \pm 2, \ldots, \pm l$
m_s	magnetic spin quantum number	$\pm \frac{1}{2}$



- Spectroscopic notation

Value of l	0	1	2	3	4	5	6
Designation	s	p	d	f	g	h	i



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Quantum Numbers – Exercise

Which of the following (n, l, m_l, m_s) combination(s) is/are impossible for an electron in an atom?

a) $(3, 1, 1, -\frac{1}{2})$

b) $(6, -2, 0, \frac{1}{2})$

c) $(3, 2, -2, -\frac{1}{2})$

d) $(3, 1, -2, \frac{1}{2})$

e) $(1, 0, 0, -\frac{1}{2})$

Quantum Numbers – Exercise

Which of the following (n, l, m_l, m_s) combination(s) is/are impossible for an electron in an atom?

a) $(3, 1, 1, -\frac{1}{2})$

b) $\boldsymbol{(6, -2, 0, \frac{1}{2})}$ **l cannot be negative**

c) $(3, 2, -2, -\frac{1}{2})$

d) $\boldsymbol{(3, 1, -2, \frac{1}{2})}$ **m_l cannot be greater than l**

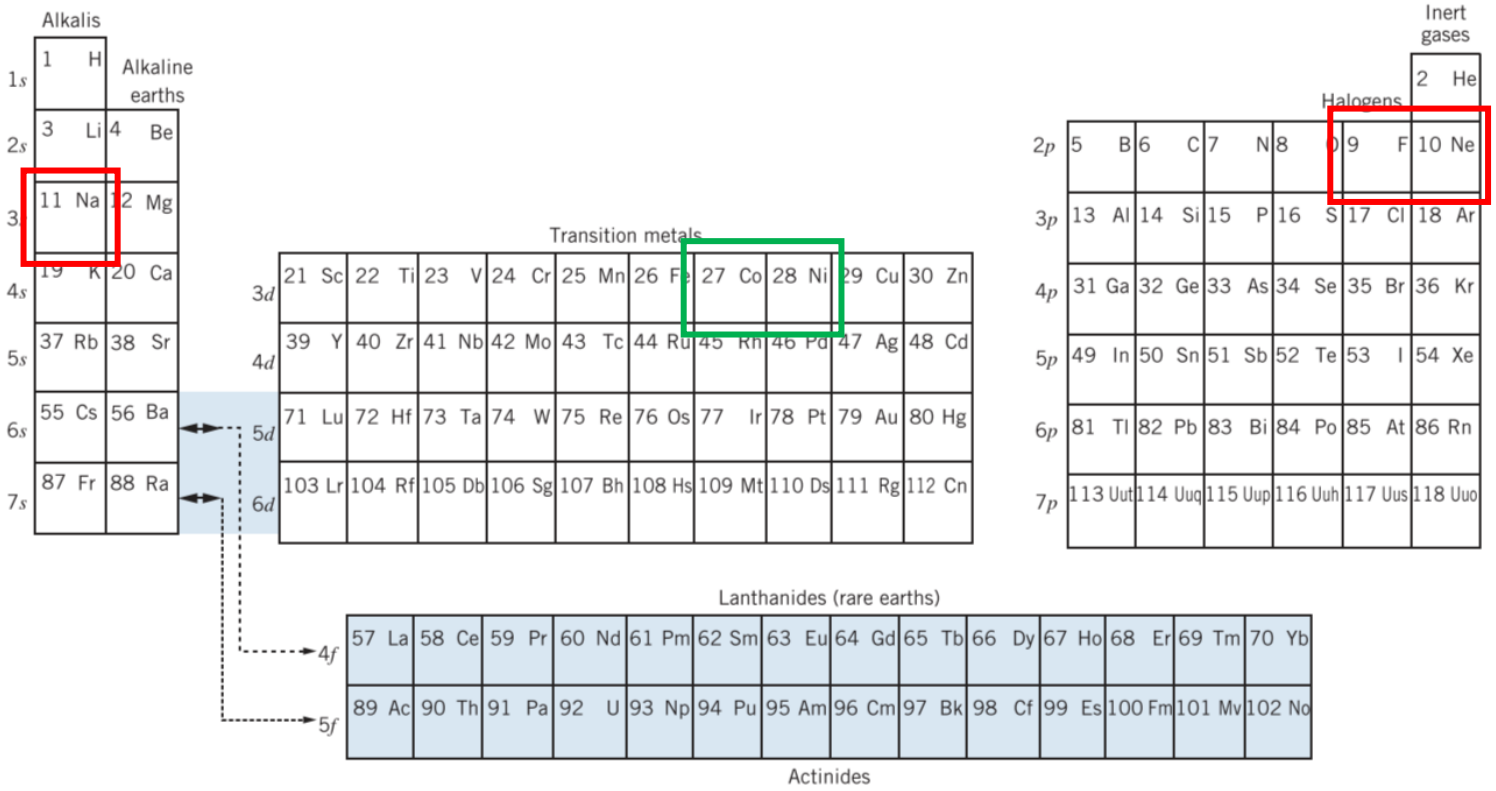
e) $(1, 0, 0, -\frac{1}{2})$

Many-electron Atoms

- Atoms have Z electrons and Z protons in the nucleus (Z →atom number)
- We cannot neglect all but one electron, since they interact with each other
  - Electrostatic forces between the electrons
- **Many-body problem:** mutual interactions of three or more objects
- The Schrödinger equation: exact, closed-form solutions cannot be found
  - Needs to be solved numerically using computers

Many-electron Atoms

- How are the energy levels occupied in an atom with Z electrons?
- If all at the lowest energy level ($1s$), we would expect smooth variations in atomic properties:



Pauli Exclusion Principle

Wolfgang Pauli
1900-1958



No two electrons in a single atom can have the same set of quantum numbers:

$$(n, l, m_l, m_s)$$

Example:

Lithium ($Z = 3$)

Three electrons need to be placed:

$$\left(1,0,0,-\frac{1}{2}\right) \rightarrow 1s$$

$$\left(1,0,0,+\frac{1}{2}\right) \rightarrow 1s$$

$$\left(2,1,0,-\frac{1}{2}\right) \rightarrow 2s$$

n	principal quantum number,	1,2,3, …
l	angular momentum quantum number,	
	magnetic quantum number,	$0, \pm1, \pm2, \ldots, \pm l$
m_s	magnetic spin quantum number	$\pm\frac{1}{2}$

Value of l	0	1	2	3	4	5	6
Designation	s	p	d	f	g	h	i

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Pauli Exclusion Principle – Exercise

A certain atom has six electrons in the $3d$ level.

(a) What is the maximum possible total m_l for the six electrons, and what is the total m_s in that configuration?

(b) What is the maximum possible total m_s for the six electrons, and what would be the largest possible total m_l in that configuration?

n	principal quantum number,	1,2,3, …
l	angular momentum quantum number,	
	magnetic quantum number,	$0, \pm 1, \pm 2, \ldots, \pm l$
m_s	magnetic spin quantum number	$\pm \frac{1}{2}$

Value of l	0	1	2	3	4	5	6
Designation	s	p	d	f	g	h	i

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Pauli Exclusion Principle – Exercise

A certain atom has six electrons in the $3d$ level.

(a) Maximum m_l:

We know $l = 2$ so, we can have $m_l = -2, -1, 0, 1, 2$. Each m_l can occupy 2 electrons ($m_s = +\frac{1}{2}$ and $m_s = -\frac{1}{2}$). To maximize m_l we start filling in the levels starting from $m_l = 2$:

- 2 in $m_l = 2$
- 2 in $m_l = 1$
- 2 in $m_l = 0$

n	principal quantum number,	1,2,3, …
l	angular momentum quantum number,	
	magnetic quantum number,	$0, \pm 1, \pm 2, \ldots, \pm l$
m_s	magnetic spin quantum number	$\pm \frac{1}{2}$

Value of l	0	1	2	3	4	5	6
Designation	s	p	d	f	g	h	i

Which makes: $2 \times 2 + 2 \times 1 + 2 \times 0 = +6$ for m_l

And: total $m_s = 0$ (equal $-\frac{1}{2}$ and $+\frac{1}{2}$)

Pauli Exclusion Principle – Exercise

A certain atom has six electrons in the $3d$ level.

(b) Maximum m_s (with largest m_l):

We know $l = 2$ so, we can have $m_l = -2, -1, 0, 1, 2$ (5 options). So, we can assign at most 5 electrons to a $m_s = +\frac{1}{2}$ state. The other one should go in a $m_s = -\frac{1}{2}$ state:

n	principal quantum number,	$1, 2, 3, \ldots$
l	angular momentum quantum number,	
	magnetic quantum number,	$0, \pm 1, \pm 2, \ldots, \pm l$
m_s	magnetic spin quantum number	$\pm \frac{1}{2}$

Which makes: $5 \times \frac{1}{2} + 1 \times -\frac{1}{2} = +2$ for m_s

Value of l	0	1	2	3	4	5	6
Designation	s	p	d	f	g	h	i

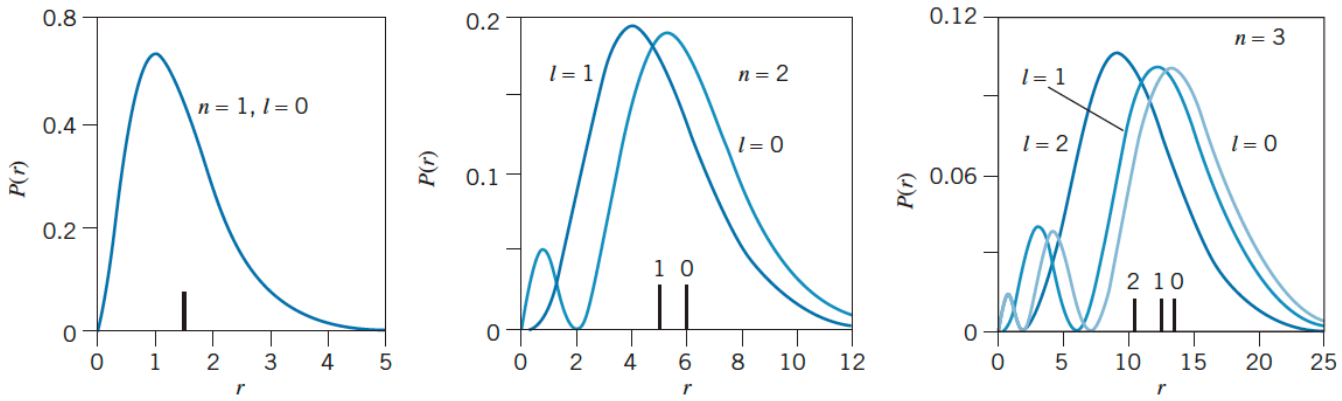
And: m_l for first 5 is 0, maximum is then $m_l = +2$

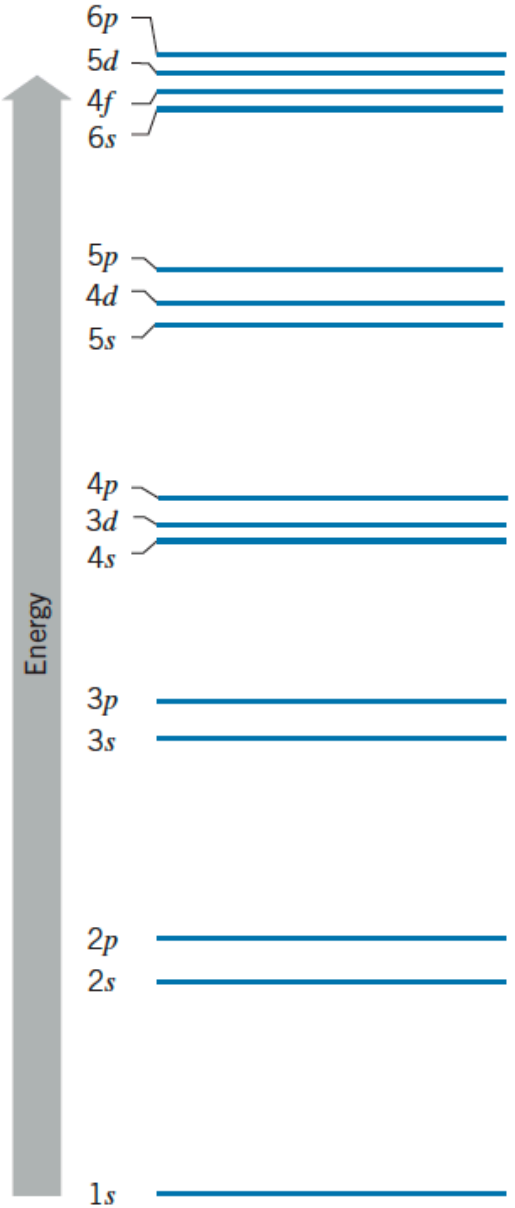
Atomic Subshells

- Atomic subshells are filled from lowest to highest energy levels

- $2s$ is energetically lower than $2p$:
  - $2s$ has greater probability to be at smaller radii

- Some levels are very close in energy levels and might differ per atom





Atomic Subshells

- The set of orbits with a certain value of n, with about the same average distance from the nucleus, is known as an **atomic shell**

n	1	2	3	4	5
Shell	K	L	M	N	O

- The levels with a certain value of n and l (for instance, $2s$ or $3d$) are known as **subshells**

- The capacity is $2(2l + 1)$ → why is that?

TABLE 8.1 Filling of Atomic Subshells

n	l	Subshell	Capacity $2(2l+1)$
1	0	1*s*	2
2	0	2*s*	2
2	1	2*p*	6
3	0	3*s*	2
3	1	3*p*	6
4	0	4*s*	2
3	2	3*d*	10
4	1	4*p*	6
5	0	5*s*	2
4	2	4*d*	10
5	1	5*p*	6
6	0	6*s*	2
4	3	4*f*	14
5	2	5*d*	10
6	1	6*p*	6
7	0	7*s*	2
5	3	5*f*	14
6	2	6*d*	10

Atomic Subshells

- Two rules for filling subshells:
  1. The capacity of each subshell is $2(2l+1)$ → Pauli excl.
  2. The electrons occupy the lowest energy state possible

- Notation for filling up the subshells: S^{n_e} with S the subshell and n_e the number of electrons in that shell

- Examples:

Hydrogen ($Z = 1$) → $1s^1$

Boron ($Z = 5$) → $1s^2 2s^2 2p^1$

Calcium ($Z = 20$) → ??

TABLE 8.1 Filling of Atomic Subshells

n	l	Subshell	Capacity $2(2l+1)$
1	0	$1s$	2
2	0	$2s$	2
2	1	$2p$	6
3	0	$3s$	2
3	1	$3p$	6
4	0	$4s$	2
3	2	$3d$	10
4	1	$4p$	6
5	0	$5s$	2
4	2	$4d$	10
5	1	$5p$	6
6	0	$6s$	2
4	3	$4f$	14
5	2	$5d$	10
6	1	$6p$	6
7	0	$7s$	2
5	3	$5f$	14
6	2	$6d$	10

Atomic Subshells

- Two rules for filling subshells:
  1. The capacity of each subshell is $2(2l+1)$ → Pauli excl.
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- Examples:

Hydrogen ($Z = 1$) → $1s^1$

Boron ($Z = 5$) → $1s^2 2s^2 2p^1$

Calcium ($Z = 20$) → $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$

TABLE 8.1 Filling of Atomic Subshells

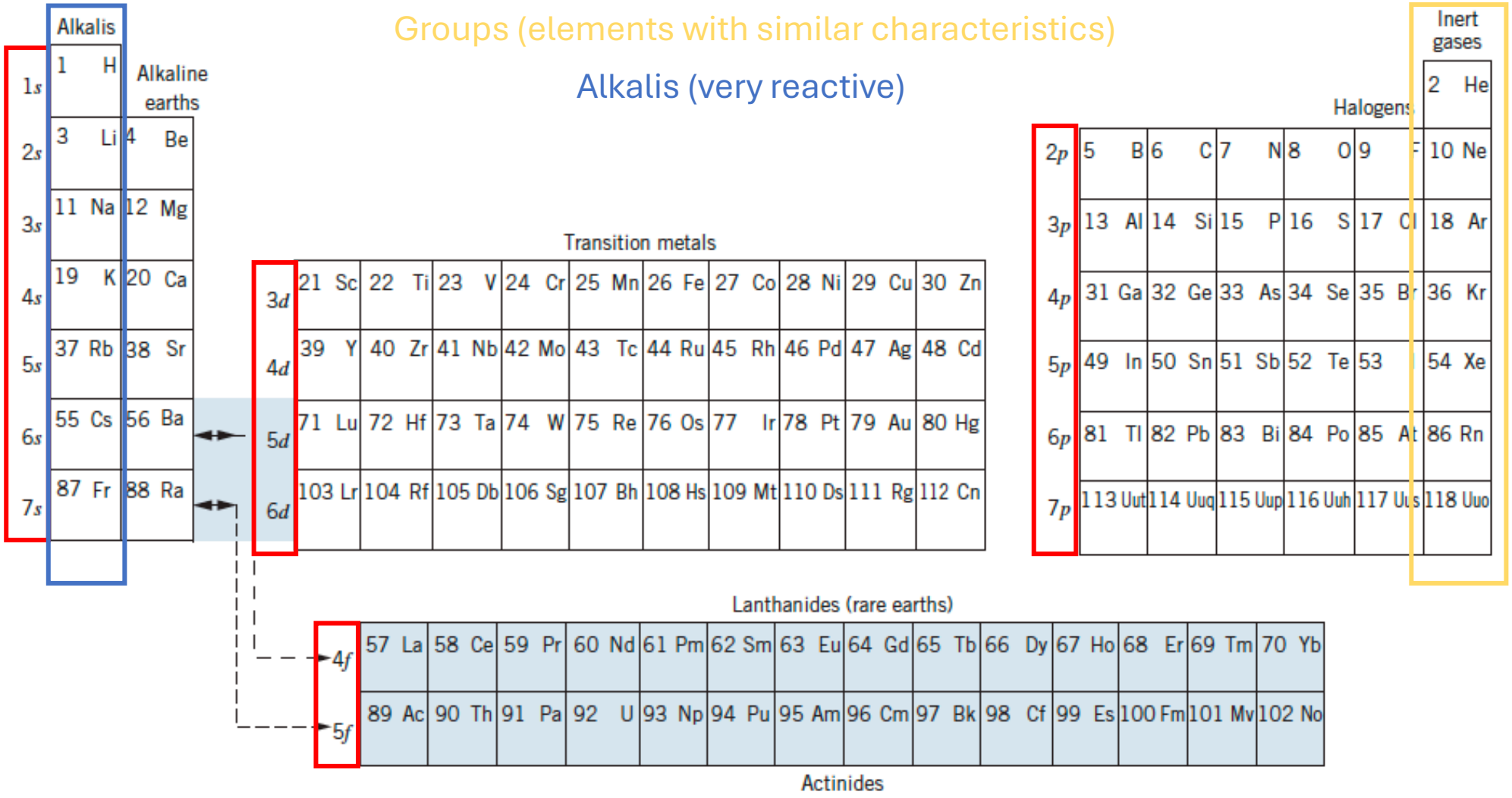
n	l	Subshell	Capacity $2(2l+1)$
1	0	$1s$	2
2	0	$2s$	2
2	1	$2p$	6
3	0	$3s$	2
3	1	$3p$	6
4	0	$4s$	2
3	2	$3d$	10
4	1	$4p$	6
5	0	$5s$	2
4	2	$4d$	10
5	1	$5p$	6
6	0	$6s$	2
4	3	$4f$	14
5	2	$5d$	10
6	1	$6p$	6
7	0	$7s$	2
5	3	$5f$	14
6	2	$6d$	10

Periodic Table

Periods (subshells)

Groups (elements with similar characteristics)

Alkalis (very reactive)



Outer Electrons

- Lithium (Li) gives up its electron very easily, but:
  - Ionization energy $\mathrm{H} = 13.6\ \mathrm{eV}$
  - Ionization energy $\mathrm{Li} = 5.39\ \mathrm{eV}$

- Based on Bohr model :

$$E_n = -\frac{m_e(Ze^2)^2}{32\pi^2\epsilon_0^2\hbar^2}\frac{1}{n^2} = -(13.60\ \mathrm{eV})\frac{Z^2}{n^2}$$

(i.e. *larger* energy expected!)

Outer Electrons

$$E_n = -(13.60\ \text{eV})\frac{Z_{eff}^2}{n^2}$$

- Inner electrons partly 'shield' part of the positive charge in the nucleus: $Z_{eff} \gtrsim 1 \rightarrow E_1 \approx -3.4\ eV$

- Additionally, the $2s$ electron can sometimes find itself much closer to the nucleus than the average radius (sometimes r<1s radius)

- *excited* state (e.g. 2p): 'higher' orbit (probability of electron near nucleus much smaller): electron screening by 1s electrons more pronounced=> 2p-electron more loosely bound

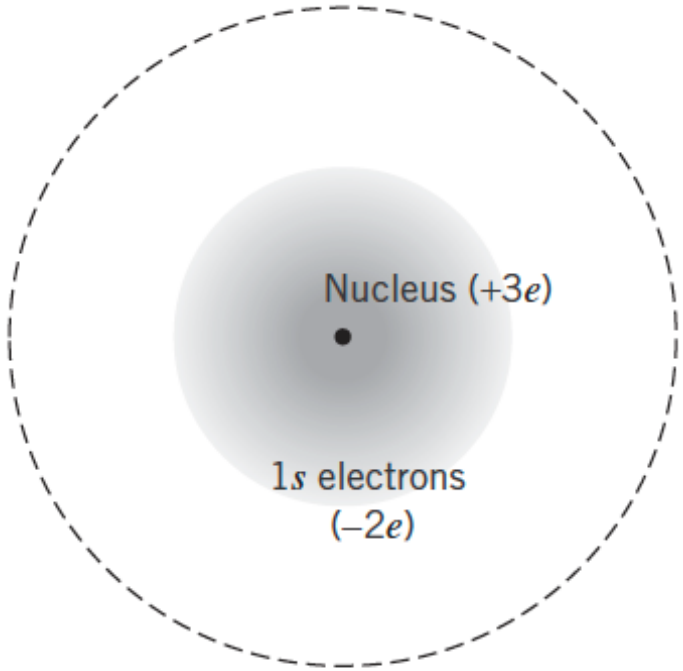


FIGURE 8.3 Electron structure in lithium, as might be seen from the average location of an outer ($2s$) electron. The dashed line represents a spherical Gaussian surface at that location.

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Outer Electrons – Example

The ground state of singly ionized lithium ($Z = 3$) is $1s^2$. Use the electron screening model to predict the energies of the $1s^1 2p^1$ and $1s^1 3d^1$ excited states in singly ionized lithium. Compare your predictions with the measured energies (respectively $-13.4\ eV$ and $-6.0\ eV$).

$$E_n = -(13.60\ \text{eV})\frac{Z_{eff}^2}{n^2}$$

- Singly ionized lithium has 2 electrons ($3 - 1 = 2$)
- If one electron is excited, it is screened by the other electron remaining in $1s$
- So: $Z_{eff} = 3 - 1 = 2$

For $2p$ ($n = 2$) electron

$$E_2 = -(13.60\ \text{eV})\frac{2^2}{2^2} = -13.60\ \text{eV}$$

For $3d$ ($n = 3$) electron

$$E_3 = -(13.60\ \text{eV})\frac{2^2}{3^2} = -6.04\ \text{eV}$$

Outer Electrons – Example

The ground state of singly ionized lithium ($Z = 3$) is $1s^2$. Use the electron screening model to predict the energies of the $1s^1 2p^1$ and $1s^1 3d^1$ excited states in singly ionized lithium. Compare your predictions with the measured energies (respectively $-13.4\ eV$ and $-6.0\ eV$).

$$E_2 = -(13.60\ \text{eV})\frac{2^2}{2^2} = -13.60\ \text{eV} \qquad E_3 = -(13.60\ \text{eV})\frac{2^2}{3^2} = -6.04\ \text{eV}$$

Note:

- Screening estimation does not consider the penetration possibility of the excited electrons into the $1s$ shell
- Estimation gets better for higher energy levels → less penetration into $1s$ → better screening effect

Ionization Energies

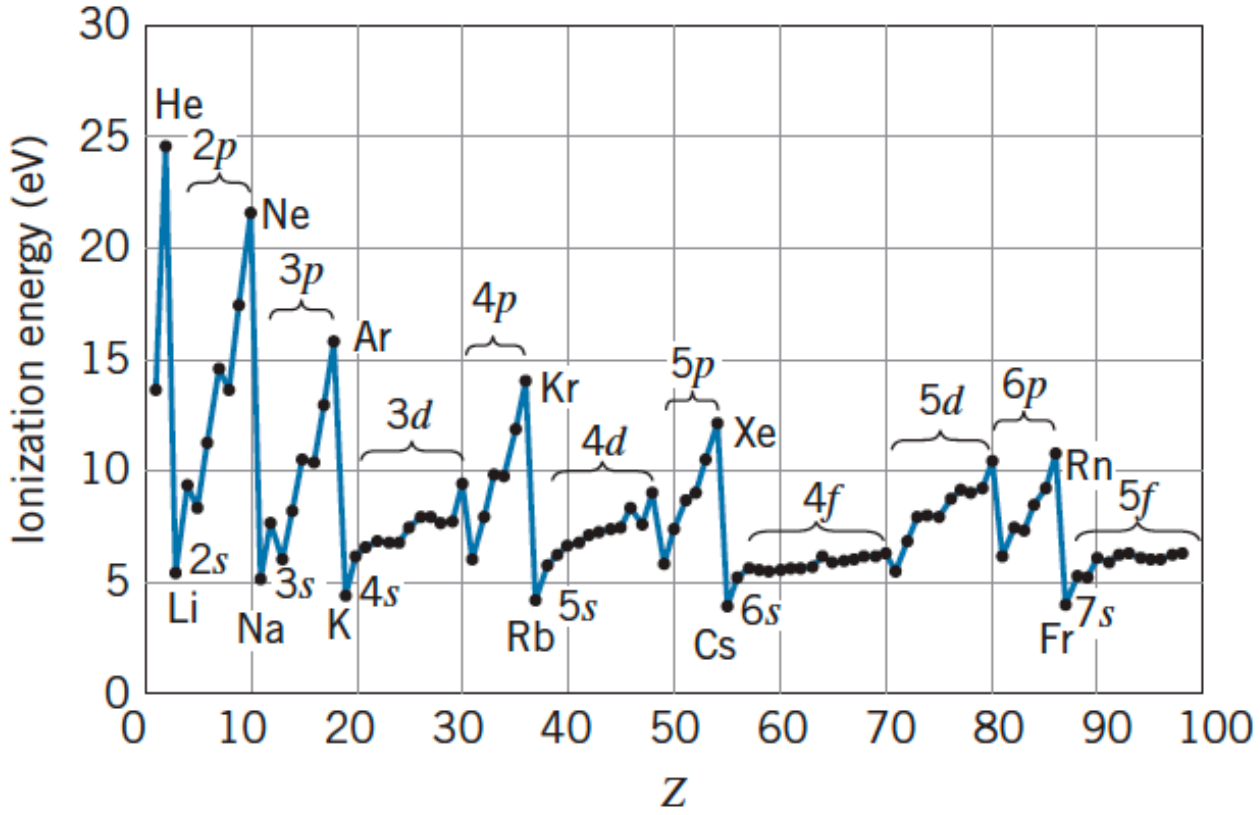


FIGURE 8.8 Ionization energies of neutral atoms of the elements.

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Optical Transitions

- We can excite outer electrons (or even ionize an atom) to create a vacancy that can be filled, releasing a photon in the process

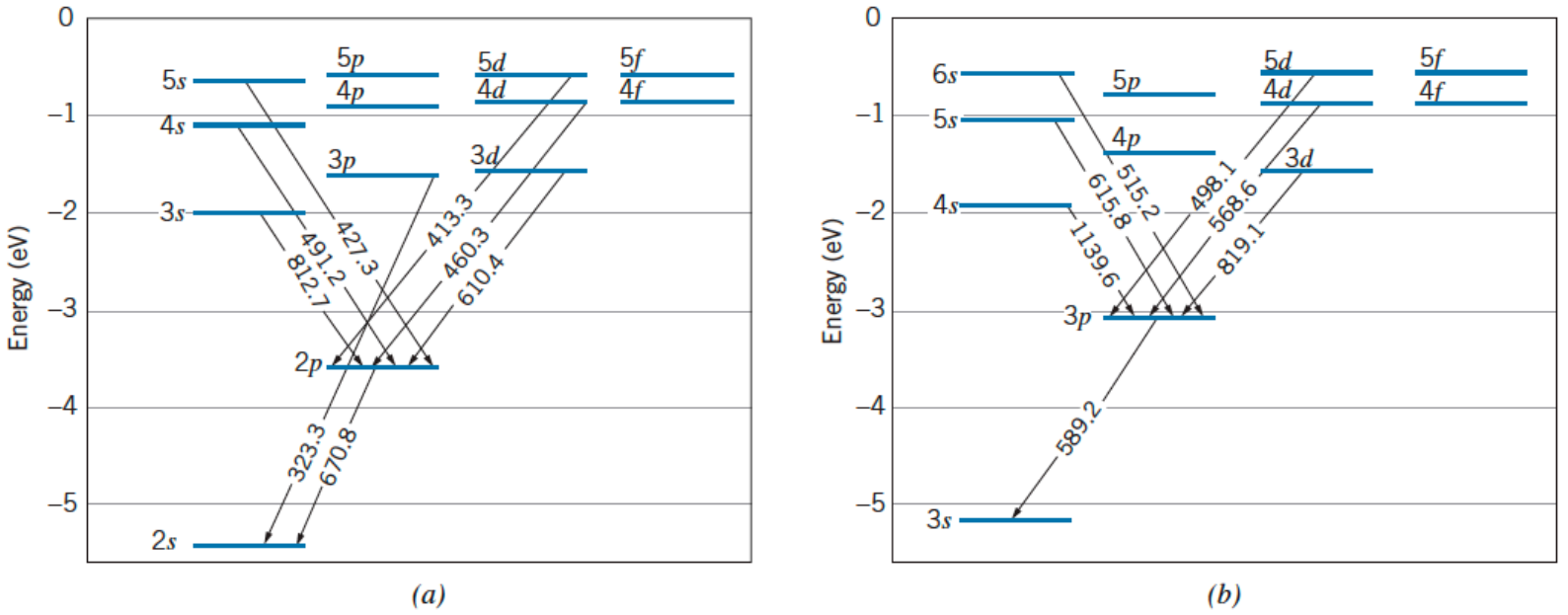


FIGURE 8.4 (*a*) Energy-level diagram of lithium, showing some of the transitions (labeled with wavelength in nm) in the optical region. (*b*) Energy-level diagram for sodium. Because of the fine-structure splitting, the 3*p* level in sodium is actually a very closely spaced pair of levels, so all transitions involving that level show two closely spaced wavelengths. The wavelength shown here is the average of the two. The fine-structure splitting is negligible for the other levels in sodium and for all of the levels in lithium.

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Optical Transitions

- The use of “black light” to cause fluorescence:



Image from: https://www.ledkia.com/blog/nl/wat-is-black-light/

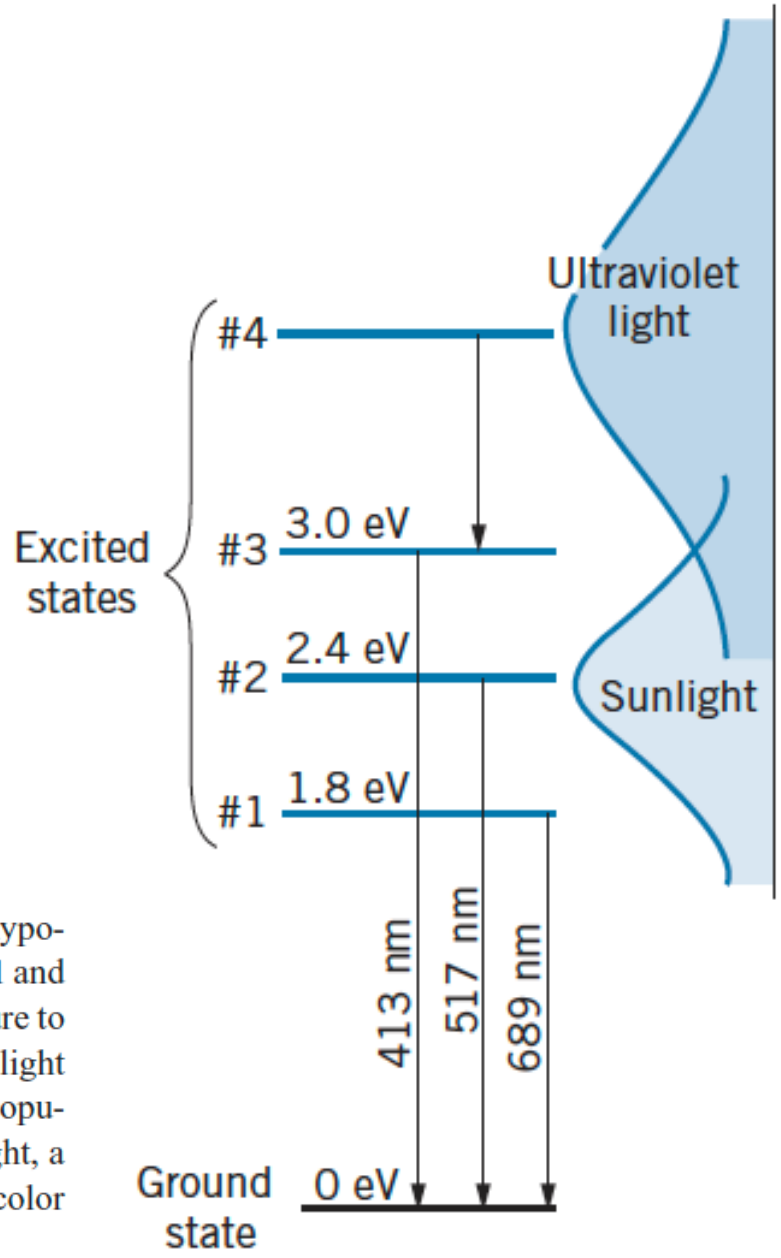
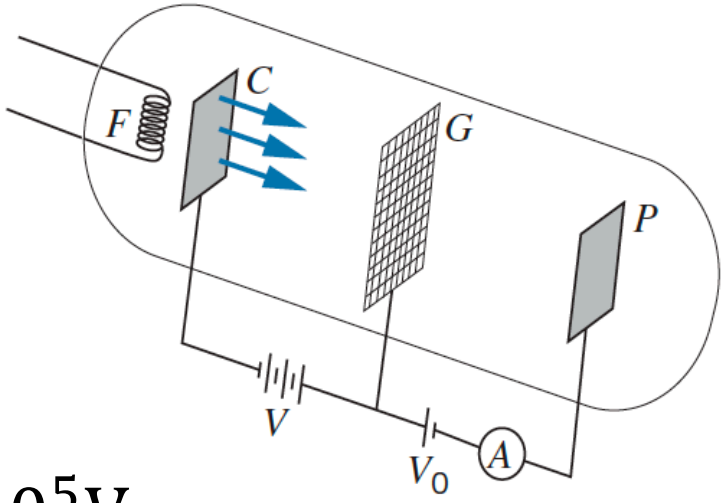
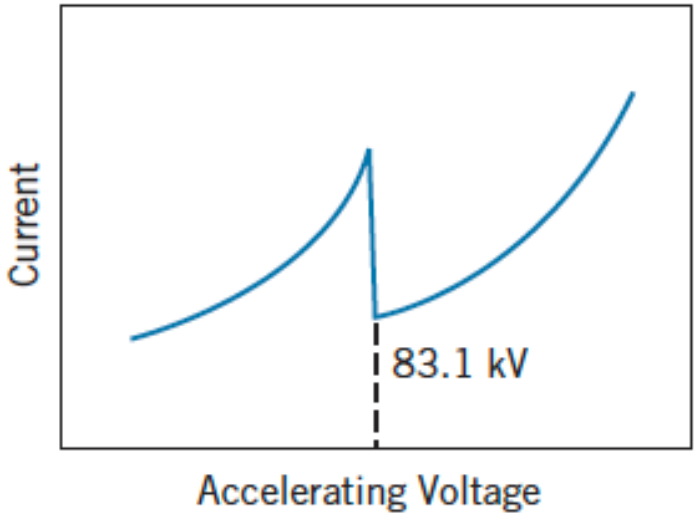


FIGURE 8.6 Excited states of a hypothetical atom. Only excited states 1 and 2 can be easily reached by exposure to sunlight; exposure to ultraviolet light populates state 4, which in turn populates state 3. Under ultraviolet light, a stronger blue or violet (413 nm) color is revealed than under sunlight.

Inner Electrons



- Recap: Franck-Hertz experiment
- Now, increase the voltage from $\sim 10\text{V}$ to $\sim 10^5\text{V}$
- Result:

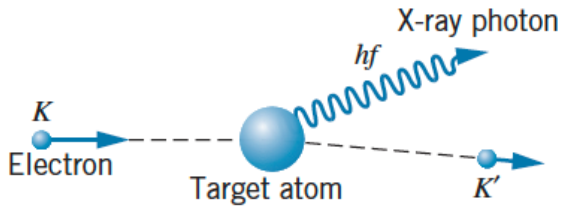


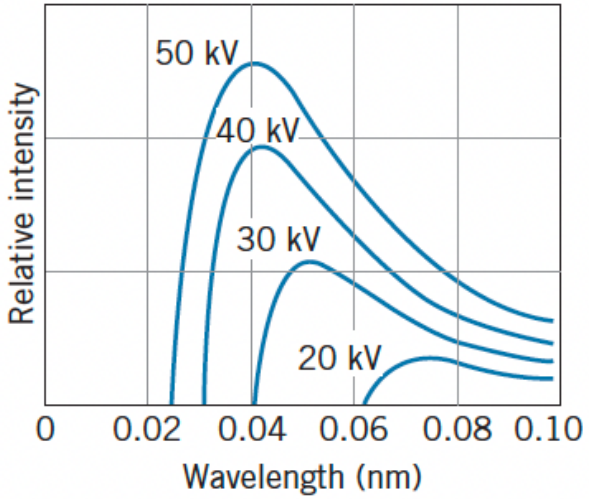
- Low accelerating voltage: push outer electrons to excited states
- High accelerating voltage: ionizing the atom by knocking loose one of the tightly bound inner electrons
- The sudden drop in the electron current is called the **absorption edge**

X-Ray Transitions

- X-Rays: EM radiation with wavelengths from 0.01 to 10 nm

- Continuous X-ray spectrum as result of *bremsstrahlung*





X-Ray Transitions

- Characteristic X-rays: photons emitted when an inner-shell vacancy is filled by a higher-energy electron

- K-series: X-rays that are emitted in the process of filling a vacancy in the K-shell
  - from $n = 2 \rightarrow K_\alpha$
  - from $n = 3 \rightarrow K_\beta$
  - from $n = 4 \rightarrow K_\gamma$
  - …

- Wavelengths/Energies depend on target material

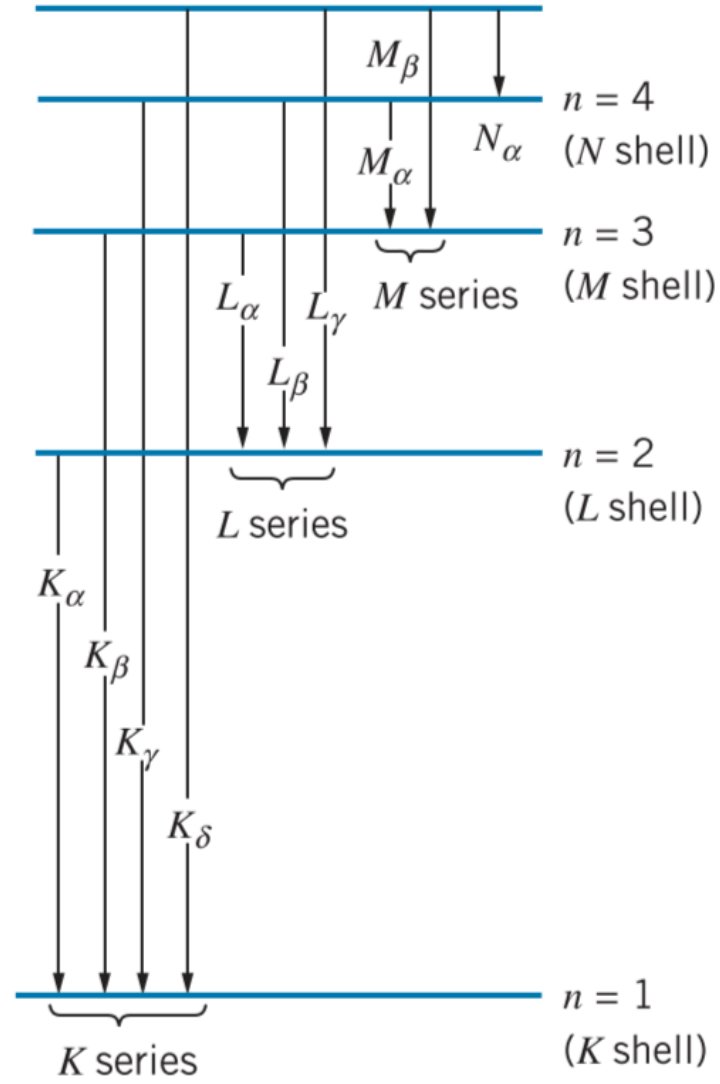


FIGURE 8.14 X-ray series.

X-Ray Transitions

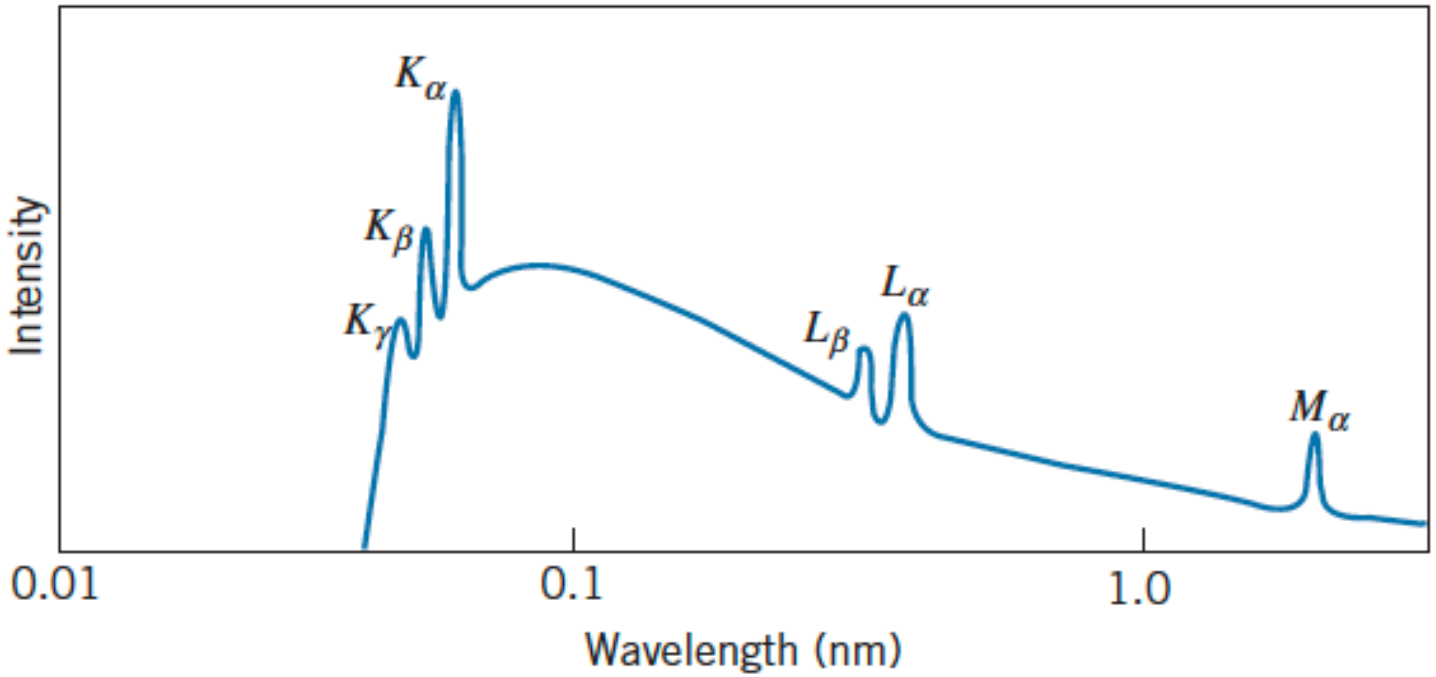


FIGURE 8.15 Characteristic X-ray spectrum of silver, such as might be produced by 30 keV electrons striking a silver target. The continuous distribution is a bremsstrahlung spectrum.